

Combinatorial & Change-Impact Analysis

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Combinatorial Analysis Background

- Premises of Combinatorial Testing
- ACTS
- Conclusion of NIST Research

Our Implementation

- How we implemented ACTS on our model constants
- How we check the results

Test Oracle Problem

- How we check the results

Data

- Statistics about degree of interaction
- How many combinatorial test cases generated, executed, passed, skipped, failed, crashed

Results

- What results we found
- What the root causes of the results were
- Bugs in our testing code
- Bugs in SPEL/LakeTemperature

Change-Impact Analysis Background

- Introduce the change-impact analysis problem
- How did specific changes to constants affect model output?

Preliminary Results

- Which constants affect which outputs
- How we did this analysis

Challenges

- Often, many model outputs change per model constant
- Output is multidimensional
- How did we pick the 3 dimensions

Example Graph

- Pick 1 constant, one output, one column, graph to time, justify decisions
- Show graph
- Explain what information it conveys

Graphing Process

- Explain how the graph was created